

*Note: Make sure to write a notebook and perform each question with proper labeling of question number
Chat GPT or any other LLM Should not be used*

Deadline : 27th December

Assignment: Handwritten Digit Classifier with PyTorch & Streamlit

Duration: 7–10 Days

Dataset: MNIST Handwritten Digits (Available directly in `torchvision.datasets`)

Problem Statement

You will build an **end-to-end machine learning project** to classify handwritten digits (0–9) using PyTorch.

You will:

1. Load & preprocess the MNIST dataset.
2. Train a **models** (CNN).
3. Evaluate the model.
4. Deploy the best model as a **Streamlit web app** where a user can draw or upload a digit image and get a prediction.
 - a. <https://docs.streamlit.io/>
 - b. https://www.youtube.com/watch?v=d7fnzDQ5qM8&ab_channel=Streamlit
 - c. https://www.youtube.com/watch?v=20V_ZB7taCM&ab_channel=JCharisTech

Tasks

Task 1 – Dataset Loading & Exploration

1. Load MNIST using:

from torchvision import datasets, transforms

2. Print:
 - Number of training & test samples
 - Image size & shape
 - First 5 sample images with their labels
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Task 2 – Data Preprocessing

1. Define **transforms**:
 - Convert image to tensor
 - Normalize to mean **0.5** and std **0.5**
 2. Create **DataLoader** objects for:
 - Training set (batch size = 64, shuffle = True)
 - Test set (batch size = 64, shuffle = False)
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Task 3 – Build Two Models

1. **Model A (CNN)**:
 - 2 convolutional layers with ReLU + MaxPooling
 - Flatten
 - Fully connected layers
 - Dropout layer
 2. Print both architectures.
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Task 4 – Training & Validation

1. Train both models for **5–10 epochs**.
2. Use **CrossEntropyLoss** and both **SGD & Adam optimizers** (try both).

3. Store:
 - Training accuracy per epoch
 - Validation accuracy per epoch
 4. Plot accuracy & loss curves.
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Task 5 – Evaluate Best Model

1. Pick the best-performing model based on test accuracy.
 2. Test it on:
 - Test set images
 - **5 custom handwritten digit images** (you can draw and save as PNG)
 3. Display predictions alongside actual labels.
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Task 6 – Save & Load Model

1. Save the model:

```
torch.save(model.state_dict(), "mnist_model.pth")
```

2. Load it in a new script to confirm predictions still work.
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Task 7 – Build Streamlit App

1. Create `app.py` that:
 - Loads the saved model
 - Lets user **upload an image of a digit**
 - Applies the same preprocessing (28×28 grayscale, normalize)
 - Displays predicted digit & probability
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Deliverables

- `training.py` → Training code for both models
 - `app.py` → Streamlit app code
 - `report.pdf` → Project report
 - Any custom images used for testing
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